RIFKY AKHMAD FERNANDA

Software Engineer | Full-Stack & Systems Architecture

Bojonegoro, Jawa Timur, Indonesia • +62-895-3350-20160 • rifkyakhmadfer@gmail.com
rifkya911.github.io • linkedin.com/in/rifkya911 • scholar.google.com/citations?user=SPRDsG4AAAAJ

PROFESSIONAL SUMMARY

Results-driven Software Engineer with 4+ years of hands-on production experience (2021–Present) architecting high-throughput retail POS systems, enterprise SaaS CRMs, AI/LLM-augmented tooling, and mission-critical government healthcare platforms. Proven track record in solving complex architectural bottlenecks: optimizing high-concurrency Oracle SQL databases by 35%, virtualizing 50K+ dataset tables to zero-stutter 60 FPS, and engineering fault-tolerant payment pipelines with 99.8% reliability. Published researcher indexed on Google Scholar and graduated Cum Laude in Informatics Engineering (GPA 3.82/4.00) with 2 registered Intellectual Property Rights (HKI).

WORK EXPERIENCE

Full-Stack Web & Android Developer, **PT Untung Bersama Sejahtera (UBS Gold)** *[Aug 2025 - Present (Surabaya)]*

- Engineered core retail Point of Sale (POS) and inventory infrastructure across national retail branches, eliminating manual checkout bottlenecks and streamlining multi-branch stock synchronization.
- Diagnosed and resolved critical database locking during peak retail trading by refactoring CodeIgniter backend logic and tuning complex Oracle SQL queries, slashing high-volume transaction latency by **35%**.
- Architected secure, high-throughput RESTful API layer connecting centralized inventory servers with Android handheld devices built in Flutter and responsive web views.
- Integrated hardware barcode scanning engine with real-time validation, mitigating input discrepancies and accelerating sales checkout speed by 40%.

Frontend Developer, **PT Tunas Kreasi Digital** *[Nov 2024 - May 2025 (Remote)]*

- Overcame severe browser rendering degradation for enterprise CRM clients managing 50K+ customer records by implementing **TanStack Virtual** and **Zustand**, achieving stutter-free 60 FPS scrolling while cutting client memory consumption by **45%**.
- Architected multi-tier data fetching with React.js and SWR, leveraging optimistic UI updates and intelligent cache revalidation to accelerate page load times by **40%**.
- Built dynamic drag-and-drop invoice builder using DnD Kit, automating customer billing workflows, client-side PDF generation, and one-click WhatsApp dispatch.
- Engineered bi-directional spreadsheet processing with ExcelJS, enabling instant bulk CRM data imports and validated template exports without server overhead.

Full-Stack Web Developer, **PT Wibu Sukses Bersama** *[Feb 2024 - Oct 2024 (Surabaya)]*

- Architected decentralized Web3 e-commerce platform on Next.js 14 and Redux Toolkit, delivering fluid sub-second navigation and high-conversion UI/UX.
- Engineered resilient fiat checkout pipeline integrating Xendit payment gateway with automated idempotent webhook handlers, ensuring zero-loss transaction processing and **99.8% payment reliability**.
- Engineered and maintained mission-critical Discord community automation bots with Node.js and Discord.js, automating member onboarding, tiered role verification, scheduled broadcasts, and server telemetry for 5,000+ members.

Full-Stack Web Developer (Freelance Agency), **PT MWTech / Daya Arsana** *[Aug 2024 - Feb 2025 (Contract)]*

- Led frontend engineering for 3 official Ministry of Health (Kemenkes RI) portals — **Balai Besar Biomedis & Genomika (BB Binomika), INA-CRC, and INA-CRR** — utilizing Next.js 14 (SSR), i18next multi-language support, and Lexical/CKEditor.
- Engineered dynamic medical questionnaire & triage builder using DnD Kit, allowing healthcare operators to configure intake schema with zero code deployments.
- Architected full-stack enterprise web portal for **Mantra Digital**, developing scalable RESTful API services in Laravel and interactive frontend interfaces with React and Radix UI components.
- Configured production reverse-proxy deployments on Nginx cloud servers, maintaining 99.9% uptime and achieving Google Lighthouse SEO/Performance scores of 95+.

EARLY PROFESSIONAL EXPERIENCE (INTERNSHIPS)

Full-Stack Web Developer (Intern), UPN Veteran Jawa Timur *[Feb 2022 - Mar 2022 (Surabaya)]*

- Engineered IBMAETER warehouse & inventory tracking web application utilizing MVC CodeIgniter 4, Bootstrap 4, and relational MySQL schemas, securing official Kemenkumham HKI registration (EC00202154284).
- Developed real-time inventory threshold push notifications via WebSockets and integrated DOMPDF for automated server-side invoice generation.

Full-Stack Web Developer (Intern), UPN Veteran Jawa Timur *[Sep 2021 - Oct 2021 (Surabaya)]*

- Designed and implemented SIM PKL student internship management portal utilizing CodeIgniter 3, relational MySQL (DDL/DML), and responsive Bootstrap, securing official Kemenkumham HKI registration (EC00202220054).
- Built authenticated RESTful API endpoints secured by stateless JWT Bearer token authentication to facilitate multi-department integration.

FEATURED SYSTEMS & AI ARCHITECTURE PROJECTS

NovaSentry — Autonomous Multi-Agent AI RAG Security Sentry | *Rust, Axum, Tokio, Vector RAG* [Repository ↗]

- Architected an asynchronous security guardrail engine in Rust that intercepts and evaluates LLM prompt injections with sub-35ms evaluation latency using local vector embeddings, paired with Model Context Protocol (MCP) server endpoints.

TapVote-AI — AI-Assisted E-Voting Kiosk & Real-Time Telemetry | *TypeScript, MCP, Mifare RFID, SSE* [Repository ↗]

- Engineered dual-ballot hardware e-voting kiosk featuring Mifare ISO 14443A RFID authentication, real-time Server-Sent Events (SSE) stream, and local Model Context Protocol (MCP) audit agents for tamper-evident tabulation.

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, PHP, Python, Golang, Rust, C++, SQL, Dart, Bash
- **AI & Automation:** Gemini API, Groq LLaMA, RAG Architecture, Model Context Protocol (MCP), Vector Embeddings, Prompt Engineering
- **Frontend & Mobile:** Next.js 15, React 19, SvelteKit, Astro, Vue, Tailwind CSS, Flutter, Radix UI, TanStack Virtual
- **Backend & APIs:** Node.js, Bun, Rust (Axum, Tokio), Laravel, CodeIgniter, Express, NestJS, REST, SSE, WebSockets, Swagger / OpenAPI
- **Databases & Cloud Storage:** PostgreSQL, Oracle SQL, MySQL, Redis, AWS S3, Cloudflare R2, MongoDB, Supabase, Prisma
- **DevOps & Observability:** Docker, Linux, Nginx, Terraform, OpenTelemetry, Grafana, Prometheus, Git, GitHub Actions, Slack

PUBLICATIONS & ACADEMIC RESEARCH (GOOGLE SCHOLAR)

- **Implementasi Metode K-Means Untuk Pengelompokan Data Kasus Covid-19 Tingkat Provinsi** — Journal research on clustering algorithms, high-dimensional data partitioning, and predictive epidemiology.
- **Studi Literatur Perbandingan Antara Metode LOC, COCOMO, dan FPA dalam Ranah Software Metric** — Research analyzing architectural software engineering estimation models and complexity metrics.
- **Sistem Pengendalian dan Pemantauan Inkubator Berbasis Web IoT dengan Metode Fuzzy Type-2** — Real-time telemetry automation and sensor control using WebSockets, microcontrollers, and fuzzy logic.

EDUCATION

Universitas Pembangunan Nasional (UPN) Veteran Jawa Timur *[Jul 2018 - Jul 2022 | Surabaya, Indonesia]*

- Bachelor of Computer Science (Informatics Engineering) — **Cum Laude (GPA: 3.82 / 4.00)** • Focus on Distributed Systems & Web Architectures
- High School Diploma in Natural Sciences (MIPA), **SMAN Model Terpadu Bojonegoro** *[Jul 2015 - Mar 2018 | Bojonegoro, Indonesia]*

HONORS, CERTIFICATIONS & INTELLECTUAL PROPERTY (HKI)

- **Intellectual Property (HKI Kemenkumham RI):** SIM PKL UPNVJ (EC00202220054) • IBMAETER Inventory System (EC00202154284)
- **Cloud & AI Certifications:** AWS Back-End Academy (Dicoding 2025) • AI Tools & API Integration for Developers (Hacktiv8 2025)
- **Professional & Honors:** Certified Web Programmer (BNSP) • Presenter SITASI 2021 • OSN Computer Science District
- **Digital Marketing & Mentorship:** SEO Virtual Internship (Niagahoster) • Learning Division Vice Chair (ILC 2018–2019)